Transformer Design for Isolated DC-DC Converter - III

Introduction

In the rapidly evolving landscape of charging infrastructure and power electronics, the efficiency and compactness of power converters are paramount. At the heart of many such systems, particularly isolated DC-DC converters, lies the transformer—a critical component responsible for voltage transformation and galvanic isolation. The design of this seemingly simple device, however, is a complex interplay of electrical, magnetic, and thermal considerations, directly impacting the overall performance, size, and cost of the converter. This lecture delves into the intricate process of transformer design, moving beyond basic principles to explore advanced optimization techniques aimed at minimizing losses and achieving optimal physical dimensions.

Understanding how to systematically design a transformer is crucial for engineers. It involves not just selecting a core and winding wires, but also making informed decisions about material properties, flux densities, and winding configurations to ensure the device operates efficiently under specific conditions. This session builds upon previous discussions, focusing on an iterative design procedure that balances competing loss mechanisms—core losses and copper losses—to arrive at a robust and optimized transformer solution for high-frequency applications.

Core Concepts

Term	Definition	Significance
`Core Loss (P_core)`	Energy dissipated in the magnetic core due to hysteresis and eddy currents, proportional to `deltaB` raised to the power `beta`.	A major source of inefficiency, its minimization is key for high-frequency operation and thermal management.
`Copper Loss (P_CU_total)`	Energy dissipated in the windings due to the resistance of the conductor, inversely	Another significant loss component, influenced by wire size, number of turns, and current density.

Term	Definition	Significance
	proportional to `deltaB` squared.	
`Peak AC Flux Density (deltaB)`	The maximum change in magnetic flux density within the core during an AC cycle.	A critical design parameter that directly influences both core and copper losses, requiring careful optimization.
`Winding Fill Factor (Ku)`	The ratio of the actual conductor area to the total window area available for windings, accounting for insulation and air gaps.	Essential for accurately calculating the available space for conductors and determining wire sizes.
`Fraction of Area Allotted (gamma j)`	The proportion of the total winding window area specifically allocated to a particular winding `j`.	Optimizing this fraction minimizes total copper losses by balancing conductor sizes across multiple windings.
`Applied Volt-Second (lambda 1)`	The integral of the voltage across a winding over a specific time period, representing the flux linkage.	A fundamental input specification derived from circuit operation, used to determine the number of turns.
`Magnetic Path Length (Lc)`	The average length of the magnetic flux path within the core material.	A core dimension that, along with cross-sectional area, influences the core's magnetic properties and inductance.
`Mean Length Per Turn (MLT)`	The average length of a single turn of wire around the core.	Used to calculate the total length of wire in a winding, which is crucial for determining winding resistance and copper loss.
`Area Product Method`	A simplified transformer design approach that uses the product of core cross-	A common starting point, but often less precise than iterative optimization methods, as it typically assumes maximum

Term	Definition	Significance
	sectional area and window area to select a core.	flux swing without loss optimization.
`Litz Wire`	A type of wire consisting of multiple thin, insulated strands twisted together.	Used to mitigate skin and proximity effects at high frequencies, reducing AC copper losses, though not explicitly covered in this simplified design procedure.

Detailed Analysis

1. Understanding Loss Mechanisms and Optimization Principles

Transformer design fundamentally revolves around managing and minimizing losses. The two primary loss components are core loss (`P_core`) and copper loss (`P_CU_total`).
- **Core Loss (`P_core`)**: This loss occurs in the magnetic core material and is given by the relationship `P_core = K_core * (deltaB)^beta` , where `K_core` is a core loss coefficient and `beta` is a material property (typically 2.5 to 2.7 for ferrite). Core loss increases with `deltaB` .
- **Copper Loss (`P_CU_total`)**: This loss occurs in the windings due to the resistance of the copper wire. It is inversely proportional to `deltaB` squared. This means as `deltaB` increases, copper loss decreases because fewer turns are needed, allowing for thicker wires.

The opposing dependencies of `P_core` and `P_CU_total` on `deltaB` imply that there is an optimal `deltaB` value where the total transformer losses (`P_total = P_core + P_CU_total`) are minimized. This optimal `deltaB` is found by differentiating `P_total` with respect to `deltaB` and setting the derivative to zero. This yields a condition where the derivative of core loss with respect to `deltaB` equals the negative derivative of copper loss with respect to `deltaB` .

Beyond `deltaB` , copper losses are also influenced by the allocation of the winding area. For a transformer with multiple windings (1 to `k`), each winding `j` is allotted a fraction of the window area, denoted as `gamma j` .
- As `gamma j` for a specific winding increases, the area available for its conductor increases, allowing for a thicker wire and thus reducing its individual copper loss.
- However, increasing `gamma j` for one winding reduces the area available for other windings, potentially forcing them to use thinner conductors and increasing their copper losses.

- Therefore, there is an optimal distribution of `gamma j` values that minimizes the total copper loss across all windings. This optimization can be achieved using advanced methods like recurrence multipliers.

2. Defining Specified Quantities and Design Outcomes

Before embarking on the design procedure, it's crucial to distinguish between quantities that are given or can be easily obtained from specifications, and those that are the outcomes of the design process.

Specified Quantities:
- **Wire Resistivity (`rho`)**: Material-dependent (e.g., 1.724 x 10^-6 ohm-cm for copper at 25°C).
- **Total RMS Winding Current (referred to primary, `I_total`)**: Estimated from circuit operation.
- **Desired Turns Ratio (`N_j / N_1`)**: Determined by nominal input/output voltages and circuit operation.
- **Applied Volt-Second of Winding One (`lambda 1`)**: Integral of `V1(t)` over the positive cycle, representing flux linkage.
- **Allowed Total Power Loss (`P_total_allowed`)**: Obtained from the overall power converter's loss budgeting and efficiency requirements.
- **Winding Fill Factor (`Ku`)**: A constant (less than 1) accounting for non-ideal packing of wires and insulation.
- **Core Loss Exponent (`beta`)**: Material property (2.5 to 2.7 for ferrites), obtained from core manufacturer data.
- **Core Loss Coefficient (`K_core`)**: Obtained from core loss curves via curve fitting.

Design Outcomes (Quantities to be Determined):
- **Core Dimensions**: Specific core size (e.g., EE25), cross-sectional area (`Ac`), window area (`WA`), mean length per turn (`MLT`), magnetic path length (`Lc`).
- **Peak AC Flux Density (`deltaB_optimum`)**: The optimized value that minimizes total losses.
- **Wire Sizes (`AW_j`)**: Cross-sectional area of the wire for each winding.
- **Number of Turns (`N_j`)**: Specific number of turns for each winding.

3. The Iterative Transformer Design Procedure

The transformer design process is iterative, meaning it involves repeated steps and adjustments to converge on an optimal solution.

Step 1: Select the Core
- Begin by selecting a candidate core, for example, an EE25 core. The goal is often to find the smallest core that meets specifications.
- From the core's datasheet, obtain its physical dimensions: `Ac` (cross-sectional

area), `Lc` (magnetic path length), `MLT` (mean length per turn), and `WA` (window area).
- Also, from the core material datasheet, obtain `beta` and `K_core`. This completes the initial set of core parameters for the first iteration.

Step 2: Evaluate Core and Copper Losses (in terms of `deltaB`)
- Express `P_core` using the selected `K_core` and `beta`, leaving `deltaB` as a variable: `P_core = K_core * (deltaB)^beta`.
- Express `P_CU_total` in terms of `deltaB`, ensuring that the `gamma j` values used are optimal for minimizing copper losses. This step assumes `gamma j` optimization has been performed or will be performed in conjunction. The expression will typically involve `rho`, `MLT`, `WA`, `Ku`, `lambda 1`, and `deltaB`.

Step 3: Optimize `deltaB`
- Formulate the total loss `P_total = P_core + P_CU_total`.
- Differentiate `P_total` with respect to `deltaB` and set the derivative to zero (`dP_total / d(deltaB) = 0`).
- Solve this equation to find the `deltaB_optimum` value. This is the peak AC flux density that minimizes the total losses for the chosen core.

Step 4: Check Total Loss Against Allowed Limit
- Substitute `deltaB_optimum` (and `gamma j_optimum`) back into the `P_total` equation to calculate the minimum total loss for the selected core.
- Compare this calculated `P_total` with the `P_total_allowed` (specified quantity).
- **If `P_total <= P_total_allowed`**: The selected core is potentially suitable. Proceed to the next step.
- **If `P_total > P_total_allowed`**: The selected core is too small or inefficient. Go back to Step 1 and select the next bigger core (e.g., if EE25 was chosen, try EE30) and repeat Steps 2, 3, and 4. This iterative process continues until a core is found where `P_total` is within the allowed limit.

Step 5: Determine Number of Turns
- Once an optimal core and `deltaB_optimum` are found, calculate the number of turns for the primary winding (`N1`) using the formula: `N1 = lambda 1 / (2 * deltaB_optimum * Ac)`.
- Then, use the specified turns ratios to calculate the number of turns for other windings: `N_j = (N_j / N_1) * N1`.

Step 6: Determine Wire Sizes
- Calculate the required cross-sectional area for each wire (`AW_j`) using the optimal `gamma j` values, `Ku`, `WA`, and `N_j`. The relationship ensures that the total area occupied by the turns of a winding fits within its allotted window space: `AW_j <= (gamma j_optimum * Ku * WA) / N_j`.
- From this calculated `AW_j`, select an appropriate standard wire gauge (AWG or

SWG) that satisfies the area requirement. The selected wire should have a cross-sectional area greater than or equal to the calculated `AW_j`.

4. Transformer Design Check and Advanced Considerations

After completing the design procedure, it's essential to perform a design check to validate the results and consider practical aspects.

Design Check:
- **Magnetizing Inductance (`LM1`)**: Calculate `LM1` (referred to the primary) using `mu_c * N1^2 * Ac / Lc`, where `mu_c` is the core permeability.
- **Peak AC Magnetizing Current (`IM_peak`)**: Calculate `IM_peak = lambda 1 / (2 * LM1)`.
- **Winding Resistances (`R_j`)**: Calculate the DC resistance for each winding: `R_j = rho * N_j * MLT / AW_j`.
- **Re-evaluate Losses**: Using the calculated `R_j` and the RMS currents, calculate the copper losses (`I_rms^2 * R_j`). Sum this with the core loss (calculated at `deltaB_optimum`) to get the total loss.
- **Final Verification**: Check if this re-evaluated total loss is still within `P_total_allowed`. If not, further iteration (e.g., selecting a larger core) might be necessary.

Advanced Considerations (Beyond this Simplified Procedure):
- **AC Effects**: The procedure primarily considers DC resistance for copper losses. At high frequencies, AC effects like skin effect (current crowding at the conductor surface) and proximity effect (current redistribution due to adjacent conductors) significantly increase effective resistance. Advanced designs incorporate these effects, often requiring Litz wire or specific winding patterns.
- **Winding Geometry**: The physical arrangement of windings (e.g., interleaved windings, number of layers) impacts leakage inductance, capacitance, and AC losses.
- **Non-Sinusoidal Excitation**: While this procedure assumes nearly sinusoidal excitation, many converters use square wave or other non-sinusoidal waveforms. More complex core loss models (e.g., improved Steinmetz equations) are needed for such cases.
- **Thermal Management**: The calculated losses generate heat. The design must ensure that the transformer can dissipate this heat without exceeding its temperature limits.

Key Takeaways

- [ ] Transformer design is an iterative process balancing core and copper losses to achieve optimal performance and size.

- [ ] Core loss increases with `deltaB`, while copper loss decreases with `deltaB`, necessitating an optimal `deltaB` for minimum total loss.
- [ ] The fraction of window area allotted to each winding (`gamma j`) must also be optimized to minimize total copper losses.
- [ ] The design procedure starts with selecting a candidate core and obtaining its physical and material properties from datasheets.
- [ ] An iterative loop compares the calculated total loss with the allowed total loss, requiring selection of a larger core if the limit is exceeded.
- [ ] Once an optimal core and `deltaB` are determined, the number of turns for each winding can be calculated using the applied volt-second and turns ratios.
- [ ] Wire sizes are determined based on the allotted window area, winding fill factor, and number of turns, followed by selection of standard wire gauges.
- [ ] A final design check involves calculating magnetizing inductance, peak current, and winding resistances to verify total losses against specifications.
- [ ] Simplified design procedures often neglect high-frequency AC effects like skin and proximity effects, which are crucial for very high-frequency applications.

Conclusion

The detailed transformer design procedure outlined in this lecture highlights the blend of scientific principles and iterative engineering required to create efficient and compact magnetic components for isolated DC-DC converters. By systematically optimizing parameters like peak AC flux density and winding area distribution, engineers can significantly reduce overall losses, leading to higher efficiency, lower operating temperatures, and smaller physical footprints—all critical factors in modern power electronics.

While this method provides a robust framework, the continuous evolution of power converter topologies and operating frequencies demands ongoing advancements in transformer design. Future considerations will increasingly focus on integrating AC loss effects, advanced winding geometries, and thermal modeling directly into the optimization process. Mastering these techniques is not just about building a component; it's about enabling the next generation of high-performance, reliable, and compact charging infrastructure and power conversion systems that underpin our electrified world.